SHAHAD ABIR AKASH

Computer Science Undergraduate | Full-Stack & ML Developer

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SUMMARY

Research-minded Computer Science undergraduate at CUET. Experienced in generative AI (Stable Diffusion, LoRA fine-tuning), NLP systems (RAG pipelines), and full-stack development with React and Node.js. Skilled in microservices architecture, CI/CD, and cloud-native deployment with Docker. Seeking software engineer or ML engineer roles.

EDUCATION

B.Sc. in Computer Science — Expected 2027
Chittagong University of Engineering & Technology (CUET) — Chittagong, Bangladesh

HSC, Science — Gov. Hazi Muhammad Mohsin College — 2021 | GPA: 5.0

SSC, Science — Khaja Ajmery High School — 2019 | GPA: 5.0

PUBLICATIONS

CYBERPUNK@DravidianLangTech 2026: Multimodal Political Meme Classification using CLIP and Logo Similarity

Shahad Abir

ACL 2026, pp. 222–226 — DOI: 10.18653/v1/2026.dravidianlangtech-1.31

- Published at the Sixth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages @ ACL 2026.
- **Rank 1** in Malayalam Subtask (F1: 0.9638) | Rank 6 in Tamil Subtask (F1: 0.9271).
- Developed a multimodal classification system combining CLIP visual embeddings, facial attribute detection, and political party logo similarity features.

SKILLS

Languages: C, C++, JavaScript, Python, Shell scripting (bash)

Frontend: React, React Native, HTML, CSS

Backend & Real-time: Node.js, Express.js, Socket.io, REST APIs

Databases: MySQL, MongoDB (NoSQL), Appwrite Database

DevOps & Microservices: Docker, Nginx, CI/CD Pipelines, Cloud Deployment, Microservices Architecture

Monitoring & Observability: Prometheus, Grafana, Elastic Stack (ELK), Kibana, Jaeger

ML / Data: NumPy, pandas, scikit-learn (classification & regression), PyTorch

Tools: Git, GitHub, Linux

Other: Problem solving, debugging, agile development

PROJECTS

Stable Diffusion Style Transfer & LoRA Fine-Tuning

PyTorch, Diffusers, PEFT LoRA, Accelerate, Hugging Face Hub

2025

- Trained LoRA adapters on Stable Diffusion v1.5 across 800 paintings from 4 classical artists (van Gogh, Matisse, Picasso, Rembrandt) with custom trigger tokens.
- Built an image-to-image stylization pipeline transferring learned classical aesthetics onto anime face inputs while preserving facial structure.
- Tuned denoising strength and cross-attention scale to achieve high-fidelity style translation with minimal structural distortion.
- Leveraged multi-GPU training via Hugging Face Accelerate for efficient fine-tuning.

Bangla Hybrid RAG Pipeline

Python, FAISS, BM25, multilingual-e5-base, BGE Reranker v2-m3

2025

- Built a Retrieval-Augmented Generation system over 169K+ Bangla Wikipedia articles (~830K overlapping chunks).
- Implemented hybrid retrieval combining BM25 lexical search and FAISS dense vector search using multilingual-e5-base embeddings.
- Improved retrieval precision through BGE Reranker v2-m3 CrossEncoder re-ranking.
- Designed Bangla-specific preprocessing including normalization, stemming, chunking, and metadata indexing.

Video Action Recognition (UCF101)

PyTorch, R(2+1)D CNN, OpenCV, 3D Convolutions

2024

- Implemented a spatiotemporal action recognition model using R(2+1)D factorized 3D convolutions on the UCF101 benchmark.
- Applied transfer learning from Kinetics-400 pretrained weights for efficient convergence.
- Built a complete video preprocessing pipeline with OpenCV for frame extraction, normalization, and temporal sampling.
- Achieved competitive classification accuracy across 101 action categories.

KrishiNexus — Agricultural Supply Chain & Climate Control Room

React, Node.js, MongoDB Atlas, Gemini 2.5 Flash, Open-Meteo API

2025

- Developed a mission-control dashboard for agricultural supply chain management and climate risk alerting across all 64 districts of Bangladesh (7th place, AI Hackathon at SciBlitz 2.0).
- Built a Three-Stage RAG advisory engine powered by Gemini 2.5 Flash and MongoDB Atlas Vector Search, injecting official BAMIS bulletins to eliminate hallucinations.
- Implemented Haversine distance-weighted logistics scoring to identify cost-optimal warehouse routes for crop dispatch.
- Created a deterministic alert engine via cron jobs cross-referencing 7-day weather forecasts against crop thresholds for dynamic district risk color-coding.

CareForAll — Microservices Donation Platform

Node.js, Express, MongoDB, Redis, Docker, Prometheus, Grafana, Jaeger, ELK

2025

- Re-engineered a legacy donation platform into a fault-tolerant microservices architecture (Top 10, CUET Micro Ops Hackathon 2025).
- Implemented Transactional Outbox Pattern and Redis-based idempotency keys ensuring at-least-once delivery and preventing double charging.
- Built CI/CD pipelines with automated testing and deployed to cloud infrastructure using Docker Compose.
- Deployed full observability stack: Prometheus metrics, Grafana dashboards, Jaeger distributed tracing, and ELK centralized logging.

CodeLens — AI Code Complexity Analyzer

MongoDB, Express, React, Node.js, Gemini API, CodeMirror, Tiptap

2025

- Built a full-stack MERN platform for code complexity analysis and developer knowledge sharing.
- Integrated Google Gemini API to analyze uploaded code and provide Big O time/space complexity estimates with structural reasoning.
- Implemented a multi-language code editor with CodeMirror supporting JavaScript, Python, Java, C, and C++.
- Developed a blogging system with Tiptap rich-text editor, JWT authentication, and admin dashboard.

RESEARCH INTERESTS

- Generative models and diffusion-based image synthesis
- Machine learning systems and model optimization
- Real-time distributed systems and WebSocket architectures

- Video understanding and representation learning
- Scalable backend and cloud-native applications

RELEVANT COURSEWORK

Core Computer Science:
Data Structures, Algorithms, Object-Oriented Programming, Database Systems, Operating Systems
Systems & Networks:
Computer Networks, Distributed Systems, Microprocessor Systems, Linux Programming
AI / Machine Learning:
Artificial Intelligence, Machine Learning, Deep Learning, Pattern Recognition
Mathematics:
Linear Algebra, Probability & Statistics, Discrete Mathematics, Numerical Methods

OPEN SOURCE CONTRIBUTIONS

- Maintained and updated multiple personal repositories on GitHub focused on backend systems and ML pipelines.
- Implemented bug fixes and feature enhancements in open-source React and Node.js projects.
- Published reusable code modules for real-time communication and data processing workflows.

ACHIEVEMENTS

1st Place — CUET Workshop Hackathon
(ft. Mysoftheaven Ltd) — Built 4 complete applications in 1 hour.

Top 10 — CUET Micro Ops Hackathon 2025
Team DevsOfOlympus — Microservices architecture competition.

Solved 300+ problems
on LeetCode and other competitive programming platforms.

Regional Olympiad Participant
in Bangladesh Physics & Math Olympiad.

EXTRACURRICULAR ACTIVITIES

IEEE Student Branch
Member, CUET Chapter.

CUET Computer Society
Active member.

Lions Club Chittagong
Contributing to community service initiatives.

LANGUAGES

Bengali
Native

English
Intermediate (B1)

Hindi
Intermediate (B1)